

PFL-ALP: Personalized Federated Learning Against Backdoor Attacks via Attention-Based Local Purification

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Abstract—Federated learning (FL) enables collaborative model training with local data privacy preserving, but is vulnerable to backdoor attacks from malicious clients. These attacks can manipulate the global model to produce malicious output when encountering specific triggers. Existing defenses, categorized as server-side and client-side approaches, have limitations such as reliance on auxiliary data availability, susceptibility to inference attacks, and instability under non-independent and identically distributed (Non-IID) data. In response to these challenges, we propose a Personalized Federated Learning via Attention-based Local Purification (PFL-ALP) algorithm, a hybrid defense mechanism integrating server-side dynamic clustering and client-side purification enhanced with personalized model knowledge. This approach effectively mitigates bias introduced by Non-IID data on the server side and further purifies the backdoored model on the client side. Specifically, we employ neural attention distillation (NAD) for model purification and enhance it with personalized model knowledge, extending the effectiveness of NAD in Non-IID FL settings. This design makes PFL-ALP compatible with privacy protocols to mitigate inference attacks. Moreover, we establish a convergence guarantee for PFL-ALP and experimentally validate its superior performance in defending against various backdoor attacks compared to multiple state-of-the-art (SOTA) defenses across three datasets. The results show that even with malicious rates ranging from 30% to 90%, PFL-ALP can reduce the attack success rate by more than 69.4 percentage points, with the reduction in main task accuracy less than 12.4 percentage points.

Index Terms—Personalized federated learning, backdoor attacks, attention maps, dynamic clustering.

I. INTRODUCTION

FEDERATED learning (FL) [1] has emerged as a powerful paradigm for collaborative model training, preserving

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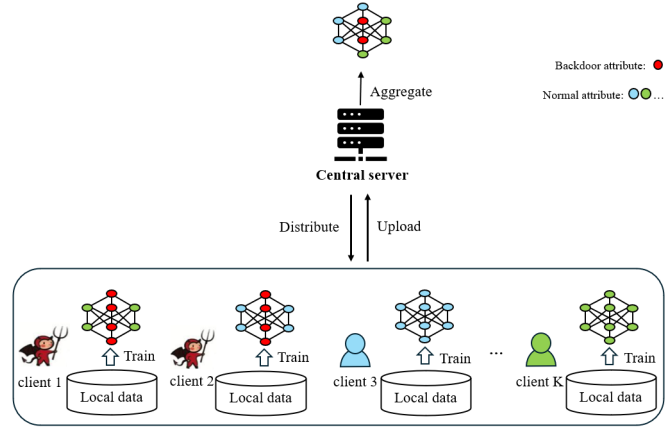


Fig. 1. A general framework of backdoor attack in FL.

data privacy by eliminating the requirement for raw data sharing. However, this distributed training paradigm exposes it to various security threats [2], [3], particularly vulnerable to backdoor attacks and inference attacks, which have become increasingly sophisticated and difficult to detect [4].

As one of the most concerning threats in FL, backdoor attacks represent a particularly insidious threat that malicious participants inject specially crafted patterns (triggers) into their local data, training their models to behave normally on most inputs but produce malicious outputs when triggers are present [5]. These triggers can take various forms, including pixel patterns, watermarks, or more subtle feature manipulations that are difficult to detect [6]. By embedding triggers into local data or directly into the training model, the attacker can implant a backdoor into the global model during the model aggregation process on the server. A general framework for such an attack is shown in Fig. 1. More sophisticated attacks involve distributed trigger mechanisms [7], where multiple attackers collaboratively embed different components of a global trigger across their local training data, making the attack more stealthy and difficult to detect.

Inference attacks [8] represent another critical security challenge in FL, aiming to reconstruct sensitive information about the training data of participants by analyzing model updates. Among them, gradient inversion attacks [9] attempt to infer training samples from gradient information, whereas membership inference attacks [10] focus on identifying whether

a given sample was used in training. The distributed nature of FL creates natural vulnerabilities to these attacks, as the shared model updates may contain implicit information about the local data of participants.

Current defense mechanisms can be broadly categorized into server-side and client-side approaches. Server-side defenses rely on the trusted federated server and typically employ statistics-based methods that leverage specific aggregation rules to remove outliers [11], [12], [13], [14]. These include techniques such as median aggregation, trimmed mean, and geometric median, which aim to filter out malicious updates based on their deviation from the distribution of benign updates.

However, current server-side defense mechanisms in FL have some significant limitations. Specifically, statistics-based methods rely heavily on statistics, which become ineffective when faced with aggressive adversaries who control a large number of clients. In such cases, the detection mechanisms can easily be overwhelmed by malicious updates. Moreover, distinguishing between benign and malicious updates becomes challenging when attackers create updates that closely resemble legitimate ones. Clipping-based approaches, in turn, are ineffective against small-magnitude malicious updates that remain within the clipping threshold. Additionally, since both statistics- and clipping-based methods require access to plaintext model updates, they are vulnerable to server-side inference attacks [9]. Therefore, most server-side strategies assume the presence of a trusted server, although this assumption may not always hold in practice. Furthermore, many server-side methods require access to clean labeled or unlabeled data for centralized validation [15], [16], which limits their applicability in the real world due to data accessibility.

Client-side defenses offer a more reliable approach by operating directly on the client side, eliminating the requirement for a trusted server or auxiliary data. These methods involve techniques such as perturbing the parameter space during local training [17], using knowledge extraction [18] and knowledge distillation [19] to mitigate the backdoor impact [20], [21]. Unlike server-side approaches, client-side defenses inherently address privacy concerns by sharing no additional information or plaintext updates with the central server. However, the local operation involved in client-side defenses, such as adding perturbations for defense, often results in performance degradation. Furthermore, our robustness analysis in Sec. V-C indicates that both server-side and client-side defenses are hindered by non-independent and identically distributed (Non-IID) data, where data across clients exhibit varying statistical properties. Similar observations are made by [22] and [23].

To overcome these challenges, we introduce the PFL-ALP algorithm, a two-step approach that combines the strengths of both server-side and client-side defenses. In the first step, we implement a server-side dynamic clustering approach that segregates model updates. Ideally, model updates are split into distinct groups: several groups consist of model updates from benign clients with similar data distributions, while one group

contains updates from malicious clients. This enables each benign client to receive a clean representative model derived from its corresponding group. However, due to the Non-IID property and sophisticated attacks designed by attackers, this ideal scenario may not always hold true. In a more realistic scenario, there would be multiple clusters of model updates, where updates within each cluster originate from both benign and malicious clients with similar data distributions. Therefore, unlike prior works that employ clustering approach for malicious detection, we adopt it exclusively as a heterogeneity mitigation tool in backdoor defense. Consequently, the representative model that each client obtained aligns with its local data distribution, but may contain malicious information. We address it in the second step on the client side.

For the design of backdoor purification, neural attention distillation (NAD) [24] has shown its effectiveness in individual machine settings, and its application has been extended to client-side defense within FL settings [20], [21]. Nevertheless, its effectiveness is limited in IID data scenario due to its reliance on local historical model knowledge, which may be skewed from global insight. When confronted with Non-IID conditions, its efficacy drops significantly [20]. To address this limitation, we equip each benign client with a personalized model that encodes more generalized knowledge. This design improves the effectiveness of NAD-based backdoor purification under Non-IID conditions and remains compatible with secure two-party computation (STPC) [25] to mitigate server-side inference risks.

The key contributions of our work are:

- 1) We propose PFL-ALP, a novel backdoor-robust FL algorithm that composes both server-side and client-side efforts. The server-side approach focuses on addressing the bias introduced by Non-IID data, and enabling benign clients to cleanse the backdoor impact with more generalized knowledge.
- 2) We extend the effectiveness of NAD into Non-IID scenarios and make it robust regardless of the number of malicious clients.
- 3) Under standard assumptions, we conduct a theoretical analysis to establish convergence guarantee for PFL-ALP.
- 4) Through extensive experiments, we show that PFL-ALP achieves superior performance against various backdoor attacks compared with multiple SOTA defense methods.

II. RELATED WORK

A. Backdoor Attacks in Federated Learning

As one of the security concerns, backdoor attacks pose significant threats to FL systems, where malicious participants inject triggers into their local training data to manipulate the output of the global model. These triggers, ranging from pixel patterns to subtle feature alterations, are challenging to detect because the local data are transparent in FL, making standard validation and visual inspection difficult to implement.

Bagdasaryan et al. [5] demonstrated how attackers could successfully backdoor FL models by uploading scaled-up

malicious updates. Xie et al. [7] introduced the distributed backdoor attack (DBA), where multiple attackers collaboratively embed different components of a global trigger across their local data, making the attack more stealthy and difficult to detect. Wang et al. [26] proposed a strategy known as “attack of the tails”, in which attackers manipulate their local models to misclassify specific minority classes while maintaining regular performance on majority classes. This can be particularly effective in FL settings where data distributions across clients may be Non-IID. Other works have explored various backdoor strategies, including flipping the training labels [27], directly poisoning model updates [28], [29], and designing stealthy triggers [6], etc.

The impact of backdoor attacks in FL is magnified by various critical factors. Firstly, the inherent invisibility of client data precludes direct inspection by the central server. Secondly, attackers can exploit the aggregation mechanism to introduce subtle but persistent patterns into the global model. Lastly, successful backdoor attacks can embed malicious behavior into a model without noticeably reducing its accuracy on the original task. This dual nature of backdoor attacks makes it difficult to detect using conventional performance monitoring methods.

B. Defense Methods Against Backdoor Attacks

Existing defense mechanisms against backdoor attacks in FL can be broadly categorized into server-side and client-side approaches.

1) *Server-Side Defenses*: Existing server-side approaches mainly rely on statistics-based outlier detection techniques to identify and filter out suspicious updates. Notably, Vucovich et al. [14] proposed FedBayes, which sets aggregation weights by contrasting the mean and variance distance between each local model and the global model, thereby suppressing malicious contributions. Flame [30] employs a cosine similarity-based HDBSCAN clustering [31] to filter out backdoored models, and further addresses backdoor impact by utilizing adaptive clipping and noising techniques based on the norm of gradients. FedDMC [32] clusters gradients using the Euclidean distance after performing dimension reduction. Multi-metrics [33] dynamically weights multiple metrics to detect backdoor attacks.

However, when facing optimization-based poisoning attacks, where malicious clients can optimize the poisoning target based on the aggregation rules, or adapt the attacks to specific metric used by the defense mechanisms, these statistical based filtering rules and robust aggregation methods may fail [4], [22], [34], [35], [36].

In addition, some strategies propose collecting primitive task-relevant data for the central server to verify the property of the received gradients. For example, Cao et al. [15] proposed FLTrust, which generates noise gradients through auxiliary data and uses cosine similarity between the received gradients and the noise gradients as weights for robust aggregation. Zhang et al. [16] assumed that there is a clean unlabeled dataset on server side, which can be labeled by the “voting strategy”. Then the backdoor can be eliminated via multi-teacher distillation.

However, these task-relevant auxiliary data-based algorithms do not consider the accessibility of private data. Although there is a vast amount of data available on the Internet, a significant portion of it is noisy and may not be suitable for the requirements of the defense algorithms.

It should be noted that most of the aforementioned server-side defense methods do not provide a privacy guarantee in the context of privacy inference attacks. Additionally, some of these methods are not compatible with current encryption approaches, such as differential privacy (DP) [37], homomorphic encryption (HE) [38] and STPC, since they necessitate access to full plaintext updates.

2) *Client-Side Defenses*: Considering the availability of auxiliary dataset and privacy constraints, client-side defense becomes a promising countermeasure since it ensures individual client security without relying on the central server. In practice, it aims to purify the backdoored model either via perturbation-based or attention-based strategies. Specifically, perturbation-based backdoor purification is generally untargeted. FL-WBC [17] perturbs the parameter space during local training to counteract the long-lasting backdoor impact, but it may introduce additional bias and degrade model performance.

By contrast, attention-based backdoor purification is targeted. It exploits the observation that backdoors hijack the attention of model toward triggers and tries to realign the attention with a benign reference. Of which, Chen et al. [39] utilize matching networks [40] to mitigate backdoor impact by removing decision logic from the backdoored model. However, its effectiveness was only tested on IID data scenarios. Utilizing the knowledge of historical clean local model, Fed-NAD [20] employs NAD to purify the global model from backdoors and is effective under IID condition. However, its effectiveness diminishes significantly in Non-IID scenarios. Similarly, leveraging the insight of local model, SPFL [21] purifies the backdoored model via attention-guided self-knowledge distillation. But its evaluation also omitted the impact of Non-IID scenarios.

Therefore, while attention-based backdoor purification is effective and addresses data availability and privacy issues inherent to server-side defenses, it remains vulnerable in Non-IID scenarios.

3) *Hybrid Defense Methods*: Some recent work has attempted to combine both server-side and client-side efforts. Among them, FLShield [41] uses representative models generated on the server side and the loss impact evaluated on the client side to validate local models prior to aggregation. Based on pivotal training, FAT [42] predicts the identities of participants on the server side and performs adversarial training on the client side. This approach enables the trained model to pivot on sensitive attributes and enhances its robustness with malicious impact. MCDFL [43] introduces a data quality detection mechanism, in which clients perform local evaluations, enabling the central server to identify and filter out malicious clients.

Despite these advances, current methods are hindered by significant constraints. Many rely on unrealistic assumptions regarding the availability of auxiliary data, the distribution of data, and the timing of attacks. Some defenses inadvertently

lead to a decline in the performance of the model on main task.

III. PROBLEM FORMULATION AND MOTIVATION

In this section, we will introduce the FL setting, the ability and goal of both attackers and defenders, as well as the motivations of PFL-ALP.

A. FL Setting

PFL-ALP focuses on the traditional horizontal FL framework, using a weighted average for aggregation. This foundational method is proposed in FedAvg algorithm [44], which can be mathematically formulated as follows:

Let θ^t be the global model in round t . Each client k employs this model to initialize its local model and performs updates using the local data D_k with Stochastic Gradient Descent (SGD). Subsequently, the updates $\nabla\ell(\theta_k^{t+1})$ are transmitted back to the server:

$$\theta_{k,e+1}^t = \theta_{k,e}^t - \eta \nabla\ell(\theta_{k,e}^t), \quad (1)$$

where e denotes the local updating epoch and η is the learning rate. Eq. (1) depicts a one-step SGD optimization process, and the final local model θ_k^{t+1} is derived from multiple local updating epochs denoted by E , i.e., $\theta_k^{t+1} = \theta_{k,E}^t$.

The server aggregates the updates using a weighted average and updates the global model:

$$\theta^{t+1} = \theta^t - \sum_{k \in S_t} \frac{|D_k|}{\sum_{k \in S_t} |D_k|} \nabla\ell(\theta_k^{t+1}), \quad (2)$$

where $|D_k|$ is the number of training samples in client k , and S_t is the selected client set. This process will be repeated for T rounds until the global model converges.

To further enhance privacy, model updates will be encrypted before sending to the server. This measure ensures that even if the server is compromised, it cannot access the raw updates from the clients. For instance, STPC prevents a single server from accessing all local updates via introducing an external server for privacy computation.

B. Threat Model

Following recent backdoor attack methods [5], [7], [26], the ability of attackers is defined as follows.

- The attacker can control a subset of clients, with the proportion of compromised clients potentially exceeds 50%.
- The attacker is able to access the local models and training data of compromised clients. They can also manipulate the local training process and craft the updated parameters sent to the server during the entire FL process.
- The attacker has complete knowledge of the aggregation algorithm and configuration, but lacks the ability to alter them.
- The attacker has no control over the benign clients whose contributions remain unaffected and genuine throughout the training process.

The primary goal of the adversary is to inject a backdoor into the global model, causing it to misclassify specific instances during inference while maintaining normal performance on the original task.

C. Nature and Goal of Defense

In PFL-ALP, the defenders include both the central server and benign clients, which are able to cooperate to suppress backdoor impact. Followed by recent backdoor defense methods [16], [21], [30], [41], we define the nature and goal of the defenders as follows:

- The defenders should not violate the privacy principle of FL, like exposing the raw data of benign clients.
- The defenders do not have the availability of additional auxiliary datasets.
- The defenders do not have knowledge about the data distribution of the participating clients.
- The defenders know nothing about the backdoor knowledge, including the proportion of malicious clients and injected backdoor triggers.
- A server is honest but curious about the raw data of clients, which requires the defense method to not violate the privacy computation mechanism.

The primary goal of defenders is to reduce the attack success rate while maintaining model accuracy on main tasks.

D. Motivation

PFL-ALP consists of two main components: dynamic clustering-based representative model generation and attention-based local purification. The motivations of PFL-ALP contain two key aspects.

Firstly, by clustering the model updates, it becomes possible to categorize clients into distinct groups based on their data distribution rather than model behavior.

Recent defense strategies [30], [41], [45] proposed the use of dynamic clustering on model updates to differentiate between benign and malicious clients. However, it becomes challenging in Non-IID scenarios, as shifts in data distribution can cause benign updates to appear more distinct than malicious ones, particularly when attackers make updates indistinguishable through dimension constraints. To be more specific, an experiment is conducted for DBA on the MNIST task under a Non-IID scenario. In this experiment, ten clients (three of which are malicious) are chosen to conduct local training and upload their model updates. The dynamic clustering results of these updates are shown in Fig. 2, with the updates projected into a 2D space through principal component analysis. The specific experimental settings are consistent with Section V-A. The results indicate that, while dynamic clustering effectively groups updates from clients with similar data distributions, it is unable to filter out backdoor updates.

Nevertheless, dynamic clustering remains valuable as it aids in grouping clients with similar data distributions into the same cluster, reducing the bias introduced by Non-IID data. By deriving a representative model from each cluster and further implementing a purification approach, it is still promising to address the impact of backdoor attacks.

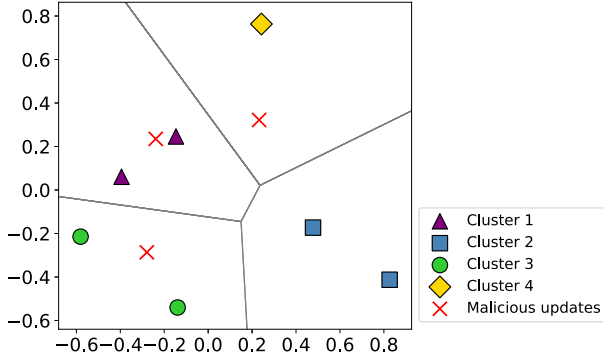


Fig. 2. A dynamic clustering result of DBA on MNIST task under Non-IID scenario.

Secondly, Nwadike et al. [46] and Li et al. [24] show that attention inconsistencies in intermediate layers between backdoored and clean models can be rectified by the knowledge of a clean model. Based on this insight, NAD has been applied to rectify backdoor information on client side in FL settings [20], [21], but it falls short in Non-IID scenarios since these methods rely on the knowledge of local historical model, which lacks generalization. To address this challenge, we employ NAD with the design of a personalized model for each benign client. The personalized model contains more generalized knowledge than the local historical model, and can be used to purify the backdoored representative model in Non-IID scenarios.

IV. PFL-ALP: PERSONALIZED FEDERATED LEARNING VIA ATTENTION-BASED LOCAL PURIFICATION

A. Overview of PFL-ALP

Before exploring the details, we first show the overall process of PFL-ALP, an overview of which is shown in Fig. 3. This framework can be split into two sides: the server side and the client side. On the server side, the focus is on identifying model updates from clients with similar data distribution, mitigating bias introduced by Non-IID data via dynamic clustering-based representative model generation. While on the client side, each benign client aims at cleansing the backdoor information embedded within the representative model, ensuring its integrity.

The pseudocode of PFL-ALP is depicted in Algorithm 1. Initially, the server broadcasts global model θ^t to each selected client (Line 3). After receiving, each client performs local update based on its individual data and sends the model gradients $\nabla\ell(\theta_k^{t+1})$ back to the server (Line 5 - 8). After receiving, the server computes the pairwise distinction among the received model gradients $\{\nabla\ell(\theta_k^{t+1})|k \in S_t\}$ and splits them into several groups $(g_1, \dots, g_c)$ via dynamic clustering (Line 10 - 14). This dynamic clustering process is instrumental in mitigating the bias introduced by Non-IID data for subsequent operations. The server then generates representative models $\{\omega_j^{t+1}|j \in [1, c]\}$ for these groups via weighted average aggregation, and updates the global model θ^{t+1} in the same way (Line 15 - 18). Subsequently, the server sends the corresponding representative model ω_j^{t+1} to each client (Line 19). After

Algorithm 1 PFL-ALP Algorithm

Input: Global model θ , Global training rounds T , Local updating epochs E , Selected client set S_t
Output: Personalized model ξ_k^{t+1} for each benign client

```

1: for each global round  $t \in [1, T]$  do
2:   //Server Executes
3:   Send current global model  $\theta^t$  to selected clients
4:   //Client Executes
5:   for each client  $k \in S_t$  do
6:     Receive global model  $\theta^t$ 
7:     Update  $\theta^t$  using local data and send gradient  $\nabla\ell(\theta_k^{t+1})$  back to the server ▷ Eq. 1
8:   end for
9:   //Server Executes
10:  Receive local updates  $\{\nabla\ell(\theta_k^{t+1})|k \in S_t\}$  from selected clients
11:  for each pair in  $\{\nabla\ell(\theta_k^{t+1})|k \in S_t\}$  do
12:     $Scores.Append(M_d(pair))$  ▷ Eq. 3
13:  end for
14:   $(g_1, \dots, g_c) \leftarrow DynamicClustering(Scores)$ 
15:  for each group  $j \in [1, c]$  do
16:     $\omega_j^{t+1} \leftarrow Generate(g_j)$  ▷ Eq. 5
17:  end for
18:   $\theta^{t+1} \leftarrow GlobalUpdate(\{\nabla\ell(\theta_k^{t+1})|k \in S_t\})$  ▷ Eq. 2
19:  Send corresponding representative model  $\omega_j^{t+1}$  to clients
20:  //Client Executes
21:  for each client  $k \in S_t$  do
22:    Receive representative model  $\omega_k^{t+1}$ 
23:    if client  $k$  is benign then
24:       $\xi_k^{t+1} \leftarrow Purification(\omega_k^{t+1})$  ▷ Alg. 2
25:    end if
26:  end for
27: end for

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receiving, each benign client performs a local purification to cleanse the representative model (Line 22 - 24). The purified representative model serves as a new personalized model, retained on the client side for the main task inference and for purification guidance in the next round. The global model serves as a warm start for local training in the next round, and its update information only serves as the foundation for dynamic clustering.

B. Dynamic Clustering-Based Representative Model Generation

To identify model updates from clients with similar data distribution, PFL-ALP designs a dynamic clustering approach based on the angular deviations of model updates.

Initially, a distinction metric is designed utilizing pairwise cosine distance to assess the angular variations across all model updates. This metric is designed to meet two key criteria. Firstly, it is designed to assign a lower value to model updates sharing akin angles and a value of zero to those with identical angles. Secondly, the metric is tailored to escalate in value in correlation with the degree of angular differences

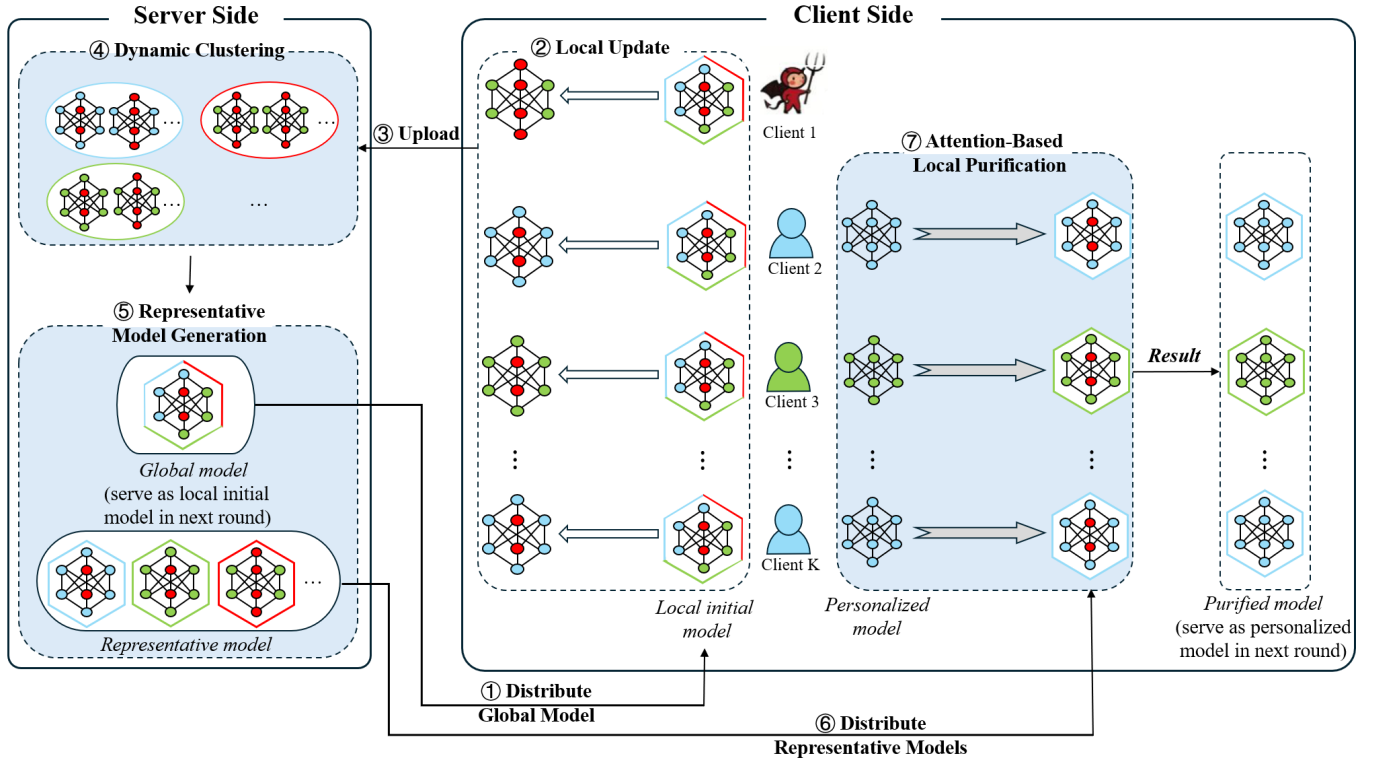


Fig. 3. Overview of proposed PFL-ALP. ① The server distributes current global model to selected clients. ② Each selected client conducts local update on the received global model. ③ After local training, clients send model updates to the server. ④ Upon receiving, the server initiates dynamic clustering, dividing model updates into several groups based on the pairwise distinction. ⑤ Subsequently, the server performs model aggregation, generating a global model and several representative models. ⑥ The server then sends the representative models to corresponding clients. ⑦ Upon receiving, each benign client performs attention-based local purification to purify the backdoored representative model, generating local purified model. This model serves as a personalized model for local inference and purification guidance in the next round. The updated global model is distributed for local update in the next round and its gradients only serve as the foundation for dynamic clustering.

between model updates. Formally, considering each pair of model updates where u represents the foundational update and v signifies the paired one, the distinction of these two updates is defined as:

$$M_d = 1 - \frac{\langle u, v \rangle}{\|u\| \|v\|}, \quad (3)$$

where $\langle \cdot, \cdot \rangle$ and $\|\cdot\|$ are inner product and Euclidean norm operators. The value of M_d ranges from 0 to 2, with lower values indicating similarity between updates and higher values reflecting increasing angular disparities.

Upon completion of the pairwise distinction computation, a comprehensive representation of the distinction values assigned to each pair of updates is obtained, denoted as matrix *scores*:

$$\text{scores} = \begin{pmatrix} s_{1,1} & s_{1,2} & \cdots & s_{1,n} \\ s_{2,1} & s_{2,2} & \cdots & s_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ s_{n,1} & s_{n,2} & \cdots & s_{n,n} \end{pmatrix}, \quad (4)$$

where $s_{u,v}$ ($u, v \in S_t$ & $|S_t| = n$) denotes the differentiation between updates u and v , and S_t signifies the chosen clients. The u -th row in the matrix *scores* illustrates the distinction values between the update u and the others.

Subsequently, the K-MEANS clustering algorithm is applied for dynamic clustering of model gradients. Given the absence of prior knowledge regarding the exact number of clustering

groups, the optimal number of clusters is determined through the evaluation of the silhouette score, a renowned metric widely acknowledged for its effectiveness in cluster assessment [47]. Upon the completion of such a dynamic clustering process, the model updates are segmented into multiple groups, with each group comprising updates from clients sharing akin data distributions. For each group g_j ($j \in [1, c]$), the server can generate the representative model through weighted average aggregation, which can be formulated as:

$$\omega_{g_j}^{t+1} = \theta' - \sum_{k \in g_j} \frac{|D_k|}{\sum_{k \in g_j} |D_k|} \nabla \ell(\theta_k^{t+1}). \quad (5)$$

To this end, we generate a representative model for each group that aggregates knowledge from clients with similar data distributions. This approach mitigates the bias stemming from the Non-IID conditions across clients. Then, to eliminate the backdoor impact, we introduce a client-side backdoor purification approach to refine the representative model.

C. Attention-Based Local Purification

Upon receiving the representative model from the server, each benign client conducts attention-based local purification to cleanse the backdoor information within the model.

As shown in Fig. 4, the attention-based local purification proceeds in two stages. Firstly, the benign client fine-tunes its

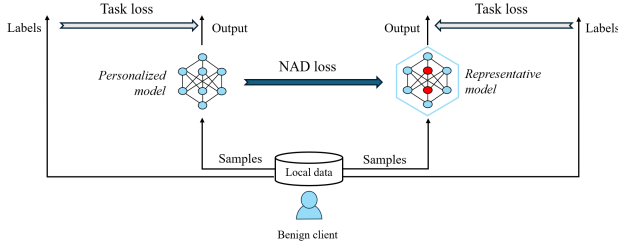


Fig. 4. Overview of attention-based local purification.

personalized model on local data to learn benign and data-consistent knowledge. Subsequently, treating this personalized model as a frozen teacher, the representative model received from the server is treated as a student. Training on the local data, the student minimizes a joint objective comprising the NAD loss and task loss that promotes attention alignment with the teacher while maintaining task performance. Then, the purified representative model is promoted as the new personalized model, which is used for inference and serves as the teacher in the next round.

Formally, let x, y be the samples and corresponding labels in local data, ξ_k be the local personalized model, using cross-entropy loss ℓ_{CE} as a criterion, the task loss in the first step can be formulated as:

$$\mathcal{L}_{CE, \xi_k} = \ell_{CE}(\xi_k(x), y). \quad (6)$$

After training, the local personalized model can be utilized to transfer benign and data-consistent knowledge. For the design of knowledge transfer, we employ NAD to align intermediate-layer attention of the backdoored model with that of the personalized model.

Formally, given the activation output at the l -th layer of the model as $A^l \in \mathbb{R}^{C \times H \times W}$, where C , H , and W indicate the channel, height, and width dimensions of the activation map respectively. A mapping function $\mathcal{F}$ operates on the activation to transform a 3D tensor into a 2D tensor across the channel dimension. The function is denoted as $\mathcal{F} : \mathbb{R}^{C \times H \times W} \rightarrow \mathbb{R}^{H \times W}$, and the formulation is given by:

$$\mathcal{F}^p(A^l) = \frac{1}{C} \sum_{i=1}^C |A_i^l|^p, \quad (7)$$

where A_i^l is the activation map of the i -th channel on l -th layer, $|\cdot|$ denotes the absolute value function. The scaling power $p > 1$ is used to amplify the disparities between the neurons of clean model and backdoored model. As suggested in [24], we set p to 2. Therefore, to cleanse the backdoored representative model using the knowledge of personalized model, we define the NAD loss at the l -th layer between them as:

$$\mathcal{L}_{NAD}(\omega_k^l, \xi_k^l) = \left\| \frac{\mathcal{F}^p(A_{\omega_k}^l)}{\|\mathcal{F}^p(A_{\omega_k}^l)\|_2} - \frac{\mathcal{F}^p(A_{\xi_k}^l)}{\|\mathcal{F}^p(A_{\xi_k}^l)\|_2} \right\|. \quad (8)$$

In addition, the representative model should also be optimized for main task accuracy, which is formulated as follows:

$$\mathcal{L}_{CE, \omega_k} = \ell_{CE}(\omega_k(x), y). \quad (9)$$

Therefore, the objective of purifying the representative model is the combination of $\mathcal{L}_{CE, \omega_k}$ and $\mathcal{L}_{NAD}(\omega_k^l, \xi_k^l)$, which is given by:

$$\mathcal{L}_{\omega_k} = \mathbb{E}_{(x,y) \in D_k} \left[\mathcal{L}_{CE, \omega_k} + \beta \sum_{l \in L} \mathcal{L}_{NAD}(\omega_k^l, \xi_k^l) \right], \quad (10)$$

where L represents the number of attention layers, β is the hyperparameter controlling the strength of knowledge transfer. Minimizing $\mathcal{L}_{\omega_k}$ not only increases the main task accuracy of representative model, but also improves the alignment of saliency maps between the personalized model and backdoored representative model.

After purification, each benign client adopts the cleansed representative model as its new local personalized model. This model is then utilized for main task inference and knowledge transfer in the subsequent purification round. Notably, in the initial training round, when benign clients have not obtained personalized models yet, they can utilize their local models as guidance for purification.

Algorithm 2 Attention-Based Local Purification

Input : Representative model ω_k^{t+1} , Purifying rounds R

Output : Updated personalized model ξ_k^{t+1}

- 1: **for** each purifying round $r \in [1, R]$ **do**
 - 2: **for** mini-batch d in D_k **do**
 - 3: Update Local personalized model ξ_k^t on main task $\triangleright$ Eq. 6
 - 4: Compute NAD loss between local personalized model ξ_k^t and representative model ω_k^{t+1} $\triangleright$ Eq. 8
 - 5: Compute main task loss for representative model ω_k^{t+1} $\triangleright$ Eq. 9
 - 6: Purify representative model ω_k^{t+1} $\triangleright$ Eq. 10
 - 7: **end for**
 - 8: **end for**
 - 9: $\xi_k^{t+1} \leftarrow \omega_k^{t+1}$
 - 10: **return** ξ_k^{t+1}
-

The process of attention-based local purification is outlined in Algorithm 2. After receiving the corresponding representative model ω_k^{t+1} , benign client k first updates the local personalized model on the main task, extracting benign knowledge from local data (Line 3). The client k then constructs a joint objective comprising the NAD loss and task loss (Line 4 - 5). By minimizing this joint objective, the representative model is purified from backdoor impact and optimized for the main task (Line 6). After purification, a clean representative model is obtained, which serves as the new local personalized model ξ_k^{t+1} (Line 9). This model is utilized for main task inference and as a guide for the next purification round.

D. Privacy Aspect of PFL-ALP

Similar to FLAME [30], FLShield [41], LSFL [48] and EPPRFL [49], PFL-ALP is compatible with STPC, which prevents the central server from accessing full local updates, thereby mitigating inference attacks. To this end, an external server can be introduced to collaborate with the central server for privacy computation. As depicted in [30] and [41], the

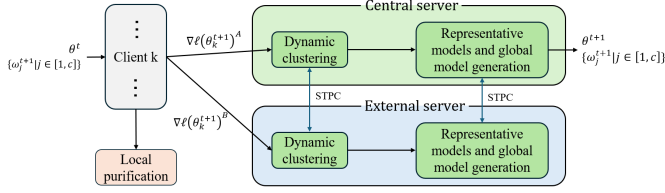


Fig. 5. Overview of PFL-ALP coupled with STPC.

cosine similarity-based distinction metric and dynamic clustering of PFL-ALP can be implemented under STPC, and the subsequent model generation is likewise compatible with this privacy-preserving framework [30], [41], [48], [49].

The key difference from existing STPC-based backdoor defense methods lies in the purpose. Prior works leverage STPC to enable secure backdoor-robust aggregation. Specifically, by securely realizing the k-nearest selection [48], or the Euclidean distance filtering and clipping [49] through STPC, the backdoor-robust aggregation can proceed without exposing raw updates. In contrast, we deploy STPC when reducing Non-IID bias by executing dynamic clustering and constructing representative and global models. The backdoor purification step, in turn, is performed locally without STPC, as it relies only on the local data and the personalized model of each benign client, making it inherently privacy-friendly.

We show the overview of PFL-ALP coupled with STPC for server-side secure computation in Fig. 5. Specifically, each selected client k trains the received global model θ^t and divides the model updates $\nabla \ell(\theta_k^{t+1})$ into two components $\nabla \ell(\theta_k^{t+1})^A$ and $\nabla \ell(\theta_k^{t+1})^B$, which are then transmitted separately to the central server and the external server. Upon receiving, the central server and the external server can jointly utilize STPC, such as the ABY framework [25] to securely perform dynamic clustering and generate representative models $\{\omega_j^{t+1} | j \in [1, c]\}$ as well as the global model θ^{t+1} . The representative models are then distributed to the corresponding clients. Subsequently, benign clients perform local purification locally on the received representative model to obtain the personalized model for inference.

The deployment of STPC ensures that a curious server cannot access to the complete local updates of clients for inference attacks, thereby enabling each benign client to obtain representative model without privacy concerns.

E. Convergence of the Local Purification

Given that the convergence of weighted average aggregation has been established in [50], we focus on the local purification phase to provide a convergence guarantee for PFL-ALP. Specifically, for a benign client k , when performing local purification, the teacher (personalized) model ξ_k is frozen and the student (representative) model ω_k is optimized via minimizing the joint objective $\mathcal{L}_{\omega_k}$ in Eq. 10. For brevity, we simplify ξ_k and ω_k as ξ and ω , $\sum_{l \in L} \mathcal{L}_{NAD}(\omega_k^l, \xi_k^l)$ as $\mathcal{L}_{NAD}(\omega, \xi)$, and $\mathcal{L}_{\omega_k}$ as $J(\omega)$. Then we make the following standard assumptions, which are prevalent in prior works [51], [52], [53].

Assumption 1: [L-Smoothness] The gradients of the two components of $J(\omega)$ are locally Lipschitz-Smooth: $\nabla \mathcal{L}_{CE, \omega}$ is L_t -smooth and $\nabla \mathcal{L}_{NAD}(\omega, \xi)$ is L_a -smooth. Consequently, the joint objective $J(\omega)$ is $L := (L_t + \beta L_a)$ -smooth.

Assumption 2: [μ -strongly convexity] $J(\omega)$ is locally μ -strongly convex around a minimizer ω^* :

$$J(\omega) \geq J(\omega^*) + \langle \nabla J(\omega^*), \omega - \omega^* \rangle + \frac{\mu}{2} \|\omega - \omega^*\|^2. \quad (11)$$

Assumption 3: [Unbiased stochastic gradients] In purifying rounds $r \in [1, R]$, let η be the learning rate, the update is $\omega_{r+1} = \omega_r - \eta g_r$ with $\mathbb{E}[g_r | \omega_r] = \nabla J(\omega_r)$ and $\mathbb{E}\|g_r - \nabla J(\omega_r)\|^2 \leq \sigma^2$.

Lemma 1: [Smoothness of $J(\omega)$]

For all ω, u in the local neighborhood, $\|\nabla J(\omega) - \nabla J(u)\| \leq L \|\omega - u\|$.

Proof: With Assumption 1 and the triangle inequality,

$$\begin{aligned} \|\nabla J(\omega) - \nabla J(u)\| &\leq \|\nabla \mathcal{L}_{CE, \omega} - \nabla \mathcal{L}_{CE, u}\| \\ &\quad + \beta \|\nabla \mathcal{L}_{NAD}(\omega, \xi) - \nabla \mathcal{L}_{NAD}(u, \xi)\| \\ &\leq L_t \|\omega - u\| + \beta L_a \|\omega - u\| \\ &= L \|\omega - u\|. \end{aligned} \quad (12)$$

□

Lemma 2: [SGD Bound under L -smoothness]

With $J(\omega)$ is L -smooth, then for the SGD step $\omega_{r+1} = \omega_r - \eta g_r$, we have

$$\begin{aligned} \mathbb{E}[J(\omega_{r+1}) | \omega_r] &\leq J(\omega_r) - \eta \left(1 - \frac{L\eta}{2}\right) \|\nabla J(\omega_r)\|^2 \\ &\quad + \frac{L\eta^2}{2} \sigma^2. \end{aligned} \quad (13)$$

Proof: With Assumption 1, we have

$$\begin{aligned} J(\omega_{r+1}) &\leq J(\omega_r) + \langle \nabla J(\omega_r), \omega_{r+1} - \omega_r \rangle \\ &\quad + \frac{L}{2} \|\omega_{r+1} - \omega_r\|^2. \end{aligned} \quad (14)$$

With $\omega_{r+1} - \omega_r = -\eta g_r$:

$$J(\omega_{r+1}) \leq J(\omega_r) - \eta \langle \nabla J(\omega_r), g_r \rangle + \frac{L\eta^2}{2} \|g_r\|^2. \quad (15)$$

With the conditional expectation on ω_r :

$$\begin{aligned} \mathbb{E}[J(\omega_{r+1}) | \omega_r] &\leq J(\omega_r) - \eta \mathbb{E}[\langle \nabla J(\omega_r), g_r \rangle | \omega_r] \\ &\quad + \frac{L\eta^2}{2} \mathbb{E}[\|g_r\|^2 | \omega_r]. \end{aligned} \quad (16)$$

Apply $\mathbb{E}[g_r | \omega_r] = \nabla J(\omega_r)$ in Assumption 3, then for $\mathbb{E}[\langle \nabla J(\omega_r), g_r \rangle | \omega_r]$ in Eq. 16 we have

$$\begin{aligned} \mathbb{E}[\langle \nabla J(\omega_r), g_r \rangle | \omega_r] &= \langle \nabla J(\omega_r), \mathbb{E}[g_r | \omega_r] \rangle \\ &= \langle \nabla J(\omega_r), \nabla J(\omega_r) \rangle \\ &= \|\nabla J(\omega_r)\|^2. \end{aligned} \quad (17)$$

Combine Eq. 16 and Eq. 17:

$$\begin{aligned} \mathbb{E}[J(\omega_{r+1}) | \omega_r] &\leq J(\omega_r) - \eta \|\nabla J(\omega_r)\|^2 \\ &\quad + \frac{L\eta^2}{2} \mathbb{E}[\|g_r\|^2 | \omega_r]. \end{aligned} \quad (18)$$

Write $g_r = \nabla J(\omega_r) + \varepsilon_r$ where $\mathbb{E}[\varepsilon_r | \omega_r] = 0$ and $\mathbb{E}\|\varepsilon_r\|^2 \leq \sigma^2$. Then for $\mathbb{E}[\|g_r\|^2 | \omega_r]$ in Eq. 18 we have

$$\mathbb{E}[\|g_r\|^2 | \omega_r] = \mathbb{E}[\|\nabla J(\omega_r) + \varepsilon_r\|^2 | \omega_r]$$

$$\begin{aligned}
&= \|\nabla J(\omega_r)\|^2 + 2 \underbrace{\langle \nabla J(\omega_r), \mathbb{E}[\varepsilon_r | \omega_r] \rangle}_{=0} \\
&\quad + \mathbb{E}[\|\varepsilon_r\|^2 | \omega_r] \\
&\leq \|\nabla J(\omega_r)\|^2 + \sigma^2.
\end{aligned} \tag{19}$$

Combine Eq. 18 and Eq. 19:

$$\begin{aligned}
\mathbb{E}[J(\omega_{r+1}) | \omega_r] &\leq J(\omega_r) - \eta \|\nabla J(\omega_r)\|^2 \\
&\quad + \frac{L\eta^2}{2} (\|\nabla J(\omega_r)\|^2 + \sigma^2) \\
&= J(\omega_r) - \eta \left(1 - \frac{L\eta}{2}\right) \|\nabla J(\omega_r)\|^2 \\
&\quad + \frac{L\eta^2}{2} \sigma^2,
\end{aligned} \tag{20}$$

which is the claimed inequality. $\square$

Lemma 3: Under μ -strongly convexity, for all ω in the neighborhood,

$$\|\nabla J(\omega)\|^2 \geq 2\mu (J(\omega) - J(\omega^*)). \tag{21}$$

Proof: Impose the first-order optimality condition, the right-hand side of Eq. 11 reaches its minimum at $\omega = \omega^* - \frac{1}{\mu} \nabla J(\omega^*)$. Then we have

$$J(\omega) \geq J(\omega^*) - \frac{1}{2\mu} \|\nabla J(\omega^*)\|^2. \tag{22}$$

Rearrange Eq. 22 yields the result in Lemma 3. $\square$

Theorem 1: [Linear convergence for the objective of local purification $J(\omega)$ w.r.t. $\mathcal{L}_{\omega_k}$]

Suppose $0 < \eta \leq 1/L$, $J(\omega)$ is bounded by

$$\mathbb{E}[J(\omega_r) - J(\omega^*)] \leq (1 - \mu\eta)^r (J(\omega_0) - J(\omega^*)) + \frac{\eta L}{2\mu} \sigma^2. \tag{23}$$

Proof: Apply Lemma 2 and take $0 < \eta \leq 1/L$, we have

$$\mathbb{E}[J(\omega_{r+1}) | \omega_r] \leq J(\omega_r) - \frac{\eta}{2} \|\nabla J(\omega_r)\|^2 + \frac{\eta}{2} \sigma^2. \tag{24}$$

Apply Lemma 3 to replace $\|\nabla J(\omega_r)\|^2$ in Eq. 24 by $2\mu (J(\omega_r) - J(\omega^*))$:

$$\mathbb{E}[J(\omega_{r+1}) - J(\omega^*) | \omega_r] \leq (1 - \mu\eta) (J(\omega_r) - J(\omega^*)) + \frac{\eta}{2} \sigma^2. \tag{25}$$

Take total expectation, unroll the recursion, and use $\mu/L \leq 1$ to write $\frac{\eta}{2} = \frac{\eta L}{2\mu} \cdot \frac{\mu}{L} \leq \frac{\eta L}{2\mu}$, the displayed bound implies the result of Theorem 1. $\square$

V. EXPERIMENTS

A. Experimental Setup

1) *Datasets and Models:* The experiments are carried out on three datasets, MNIST [54], Fashion-MNIST [55], and CIFAR-10 [56]. In order to effectively simulate the attacks, we adopt the same model architectures as used in previous attack studies. Specifically, for the MNIST handwritten recognition task and the Fashion-MNIST commodity recognition task, we implement the convolutional neural network introduced in [7]. For the CIFAR-10 image recognition task, we use the same ResNet18 architecture presented in [5].

2) *Backdoor Attacks:* We evaluate PFL-ALP against five prominent attacks: Semantic Attack (SA) [5], Edge-Case Backdoor Attack (ECBA) [26], Badnet Attack (BA) [6], DBA [7] and Targeted Label Flipping Attack (TLFA) [43], all of which are well-established and proven to be effective.

- 1) *SA.* We evaluate the defense methods against all three trigger types: background wall (wall), racing stripe (stripe) and green car (car).
- 2) *ECBA.* We follow the settings in [26], where the attacker utilizes images from Southwest Airlines' plane class as edgecase samples on CIFAR-10 and mislabels them as trucks.
- 3) *BA.* We implement it on the CIFAR-10 task, the attacker inserts a pixel block as backdoor triggers of varying sizes: 3×3 , 5×5 , and 7×7 .
- 4) *DBA.* We apply this attack across all three datasets using trigger patterns consistent with [7], which are split into 4 parts.
- 5) *TLFA.* This attack is a targeted poisoning attack with objectives akin to backdoor attacks, aiming to compromise the training model to misclassify specific samples into a target class while maintaining overall model performance on other samples. So we also test the defense methods against this attack. Following the settings in [43], the class pair for CIFAR10, Fashion-MNIST and MNIST are (automobile, truck), (coat, shirt), (digit 5, digit 3).

3) *Baseline Defenses:* We compare the performance of PFL-ALP with FedAvg [44] (aggregation without any defense) and five SOTA defenses, including three server-side defenses: FLTrust [15], FLAME [30] and Multi-metrics [33], a client-side defense: FL-WBC [17], and a hybrid defense: FLShield [41]. These methods have been introduced in Section II.

4) *Data Distribution:* We consider both IID and Non-IID scenarios and follow the one class-expert distribution setting introduced in [15], [41]. Specifically, clients are randomly divided into c groups, where c corresponds to the number of classes in the dataset. For all three datasets in this experiment, c is 10. Training samples with label j are allocated to group j with a probability of p , where $0 < p < 1$, and to any other groups with a probability of $\frac{1-p}{c-1}$. In the IID scenario, we set p to 0.1, ensuring that each sample is assigned to a group with an equal probability of 10% while for the Non-IID scenario, we set p to 0.5. In this setting, samples with label j have a 50% likelihood of being assigned to group j , and a probability of $\frac{1-0.5}{10-1} = 5.6\%$ to be allocated to other groups.

5) *Parameter Settings:* The experiment is conducted with a total of $K = 100$ clients, and unless specified otherwise, we maintain the percentage of malicious clients as 30%. In each training round, we select 10 clients, three of which are malicious. The global training process spans 100 rounds. During the training process, benign clients perform $E = 1$ epoch of local training with batch size $B = 64$ and learning rate $\eta = 0.1$; while adversaries implement attacks with $E = 6$, $B = 64$ and $\eta = 0.05$. The adversaries implement attacks consistently throughout the FL training process.

TABLE I
ROBUSTNESS COMPARISON OF DEFENSES AGAINST VARIOUS CHALLENGING ATTACKS ON CIFAR-10 IN IID SCENARIO (%)

Defense	Wall		SA Stripe		Car		3 × 3		BA 5 × 5		7 × 7		ECBA		RS↑
	MA↑	ASR↓	MA↑	ASR↓	MA↑	ASR↓	MA↑	ASR↓	MA↑	ASR↓	MA↑	ASR↓	MA↑	ASR↓	
FedAvg [44]	73.61	100	70.92	100	<u>72.92</u>	100	81.92	88.23	<u>82.75</u>	95.03	83.70	96.62	61.78	86.22	0
FLTrust [15]	71.90	6.67	71.23	20.00	66.83	3.33	68.87	<u>3.92</u>	<u>67.75</u>	<u>5.66</u>	63.90	6.18	58.48	11.73	<u>5.66</u>
FL-WBC [17]	63.02	<u>2.10</u>	62.96	<u>1.43</u>	63.75	<u>0.81</u>	80.07	92.76	80.16	<u>96.21</u>	80.61	98.07	57.71	39.21	2.88
FLAME [30]	70.69	16.67	<u>73.23</u>	13.33	71.41	27.33	67.05	5.06	64.36	5.76	66.17	<u>5.83</u>	<u>64.64</u>	<u>7.65</u>	5.56
Multi-metrics [33]	59.64	0.00	67.54	0.00	62.19	3.33	68.72	32.71	65.25	35.17	68.96	62.00	45.92	48.53	3.83
FLShield [41]	66.87	0.00	69.27	0.00	71.1	0.00	<u>81.77</u>	90.45	82.78	94.43	<u>82.07</u>	95.28	58.99	100	2.63
PFL-ALP	<u>73.45</u>	0.00	73.57	0.00	73.30	0.00	70.83	3.45	71.15	4.99	72.38	3.65	65.97	2.03	6.54

The **best** and second-best results are highlighted in bold and underline respectively.

TABLE II
ROBUSTNESS COMPARISON OF DEFENSES AGAINST DBA AND TLFA ON VARIOUS DATASETS IN IID SCENARIO (%)

Defense	MNIST				Fashion-MNIST				CIFAR-10				RS↑
	DBA		TLFA		DBA		TLFA		DBA		TLFA		
	MA↑	ASR↓	MA↑	ASR↓	MA↑	ASR↓	MA↑	ASR↓	MA↑	ASR↓	MA↑	ASR↓	
FedAvg [44]	99.04	99.93	92.16	76.68	88.40	99.98	81.77	80.40	73.69	98.27	62.46	72.10	0
FLTrust [15]	98.36	53.94	97.03	1.46	79.82	9.86	85.01	10.94	60.74	<u>39.59</u>	65.06	25.32	<u>4.31</u>
FL-WBC [17]	97.96	99.96	96.98	1.68	84.79	99.92	77.74	16.13	57.82	70.94	56.99	77.55	1.63
FLAME [30]	96.72	0.53	96.66	4.80	79.18	<u>9.67</u>	84.13	27.54	64.98	98.03	<u>66.12</u>	25.90	4.03
Multi-metrics [33]	<u>98.97</u>	99.93	<u>97.91</u>	<u>1.01</u>	<u>85.58</u>	99.98	<u>85.61</u>	16.3	<u>67.65</u>	43.71	47.16	<u>11.36</u>	2.93
FLShield [41]	97.19	4.20	97.11	2.69	78.89	10.48	85.18	<u>6.40</u>	62.34	99.74	63.76	19.18	4.29
PFL-ALP	95.33	<u>3.35</u>	98.84	0.45	81.93	8.04	85.75	5.04	62.33	13.86	71.94	11.11	5.53

The **best** and second-best results are highlighted in bold and underline respectively.

6) *Performance Metrics*: We evaluate the performance with three metrics, including main task accuracy (MA), attack success rate (ASR) and ranking score (RS). MA assesses the accuracy of the model in the main task, and ASR quantifies the accuracy of the model on backdoored data. RS is the overall improvement across various attacks compared to FedAvg. Specifically, following [33], [57], we assess the improvement over FedAvg via $\frac{MA_D - MA_B - ASR_D - ASR_B}{MA_B \cdot ASR_B}$, where MA_D and ASR_D are the MA and ASR of defense method, MA_B and ASR_B are those under FedAvg. Summing the results over various attacks yields the RS.

For defenses based on personalized model, like PFL-ALP and FL-WBC, MA and ASR are reported as the mean across personalized models of benign clients, while for other single global model defense, these metrics are reported as the performance of the final global model.

7) *Hardware Environment*: Our experiments are conducted using PyTorch 2.4.1 on a Ubuntu 24.04 system equipped with an RTX GPU 8000e and an Intel(R) Xeon(R) Gold 5218 processor running at 2.30GHz.

B. Performance Comparison With SOTA

Tables I, II, III and IV show the overall comparison of PFL-ALP with FedAvg (no defense) and five SOTA defenses against various attacks on three datasets in both IID and Non-IID scenarios.

Specifically, when countering SA, BA and ECBA on CIFAR-10 dataset, as shown in Tables I and III, PFL-ALP con-

sistently achieves the best overall performance and robustness under both IID and Non-IID conditions. Specifically, in the IID scenario, the defense performance of FLTrust (the second-best) and FLAME (the third-best) fluctuates, especially under SA across different trigger types. In contrast, PFL-ALP continuously achieves both the highest MA and the lowest ASR in most attack scenarios. For example, when defending against SA, PFL-ALP maintains robustness in suppressing backdoor effects with ASR of 0%, while maintaining competitive MA around 73%. Therefore, it achieves the highest RS of 6.54.

Under the more challenging Non-IID setting, the efficacy of most defense methods suffers substantial degradation. For example, FLTrust and FLAME still effectively suppress ASR, but at the expense of severe MA drops, which often fall below 50%. FL-WBC and FLShield do not maintain robustness, showing an ASR higher than 90% under BA. In contrast, PFL-ALP preserves both high MA and low ASR. Its MA remains above 65% for semantic attack across all three triggers, above 61% for BA across various trigger size and the highest for ECBA, with ASR remaining near 0 – 5%. This stability leads to the best overall RS as 6.57, confirming that PFL-ALP effectively mitigates the backdoor impact and generalizes well to heterogeneous data.

Similar results are shown in DBA and TLFA across three datasets in Tables II and IV. Specifically, in IID setting, PFL-ALP excels in most scenarios, except defending against DBA on MNIST task, in which FLAME achieves the best performance with MA of 96.72% and ASR of 0.53%, while

TABLE III
ROBUSTNESS COMPARISON OF DEFENSES AGAINST VARIOUS CHALLENGING ATTACKS ON CIFAR-10 IN NON-IID SCENARIO (%)

Defense	Wall		SA Racing Stripe		Green Car		3 × 3		BA 5 × 5		7 × 7		ECBA		RS↑
	MA↑	ASR↓	MA↑	ASR↓	MA↑	ASR↓	MA↑	ASR↓	MA↑	ASR↓	MA↑	ASR↓	MA↑	ASR↓	
FedAvg [44]	62.94	100	56.21	86.67	58.87	100	76.05	89.93	78.24	91.31	77.73	95.08	53.13	62.25	0
FLTrust [15]	51.65	0.00	55.71	0.00	47.90	0.00	39.78	10.05	39.86	<u>6.06</u>	39.41	<u>12.92</u>	46.36	<u>13.37</u>	<u>4.51</u>
FL-WBC [17]	43.72	<u>10.00</u>	46.01	<u>1.05</u>	48.69	0.90	<u>74.56</u>	91.63	73.70	95.21	73.18	96.57	44.91	49.57	2.05
FLAME [30]	41.80	0.00	42.07	2.67	41.78	<u>0.67</u>	45.20	<u>4.32</u>	48.59	10.65	53.41	76.27	35.30	21.43	3.34
Multi-metrics [33]	34.54	0.00	35.57	5.33	38.27	9.33	67.25	<u>67.95</u>	67.88	90.94	67.21	93.90	30.68	99.48	0.53
FLShield [41]	47.30	0.00	51.97	1.33	54.88	20.67	73.96	92.77	<u>73.96</u>	92.76	<u>73.77</u>	97.87	31.21	14.29	2.53
PFL-ALP	65.48	0.00	66.54	0.00	66.15	0.00	61.40	3.83	61.69	5.43	61.86	4.19	53.84	2.41	6.57

The **best** and second-best results are highlighted in bold and underline respectively.

TABLE IV
ROBUSTNESS COMPARISON OF DEFENSES AGAINST DBA AND TLFA ON VARIOUS DATASETS IN NON-IID SCENARIO (%)

Defense	MNIST				Fashion-MNIST				CIFAR-10				RS↑
	DBA		TLFA		DBA		TLFA		DBA		TLFA		
	MA↑	ASR↓	MA↑	ASR↓	MA↑	ASR↓	MA↑	ASR↓	MA↑	ASR↓	MA↑	ASR↓	
FedAvg [44]	98.94	99.93	90.43	96.86	87.97	99.98	<u>84.18</u>	52.80	<u>54.3</u>	91.21	<u>54.46</u>	92.30	0
FLTrust [15]	97.81	99.64	98.13	0.56	76.64	<u>15.61</u>	77.19	10.84	<u>37.40</u>	<u>9.09</u>	36.69	32.32	<u>3.41</u>
FL-WBC [17]	86.92	<u>97.07</u>	88.75	5.57	80.32	99.97	79.72	23.09	39.38	11.28	44.65	70.65	1.91
FLAME [30]	98.48	99.93	98.59	<u>0.45</u>	75.00	97.96	78.12	28.14	35.24	54.20	34.33	<u>14.22</u>	1.88
Multi-metrics [33]	98.23	99.86	97.95	1.23	78.15	99.98	75.28	19.62	33.04	94.59	23.18	17.26	1.29
FLShield [41]	<u>98.49</u>	99.98	<u>98.69</u>	0.67	<u>84.78</u>	99.98	76.03	<u>3.96</u>	59.35	98.51	31.12	25.10	2.18
PFL-ALP	94.06	6.08	98.73	0.40	70.86	11.62	86.26	3.73	41.06	6.77	61.18	8.56	5.33

The **best** and second-best results are highlighted in bold and underline respectively.

PFL-ALP remains the second-best with MA of 95.33% and ASR of 3.35%. However, when it comes to the Non-IID scenario, FLAME and other defenses fail in this task, with ASR exceeding 97%, while PFL-ALP remains robust with an ASR of 6.08%.

Furthermore, it is noticed that other competing methods also encounter challenges in certain attacks under Non-IID condition, despite their efficacy in IID scenario. For instance, Multi-metrics demonstrates an ASR of 32.71% and 35.17% for BA with trigger sizes of 3×3 and 5×5 on CIFAR-10 in the IID scenario, yet these rates escalate to 67.95% and 90.94% in the Non-IID setting.

Defenses like Flame, FLTrust, and FLShield employ dynamic clustering on local updates for Non-IID heterogeneity mitigation or backdoor identification, and further leverage clipping, filtering and noising techniques to eliminate malicious information. However, these defense methods can inadvertently introduce side effects in IID scenarios. For example, when defending against DBA in the CIFAR-10 task, FLTrust achieves an ASR of 9.09% in the Non-IID scenario, but this rate increases to 39.59% in the IID scenario. Similarly, to defend against TLFA in the CIFAR-10 task, Flame achieves an ASR of 14.22% in the Non-IID scenario, which rises to 25.90% in the IID scenario. Similar situation also shown in the results of FLShield, which is designed to counter attacks occurring during the middle rounds of FL training, struggles in defending against attacks throughout the entire process. In

contrast, PFL-ALP maintains robustness in both IID and Non-IID scenarios due to its local purification technique based on personalized model knowledge, which emphasizes generalized knowledge over introducing additional noise. As a result, the RS of PFL-ALP is both the most robust across IID and Non-IID settings and the highest overall, exceeding FLTrust (the second-best) by about 1 point in IID setting and about 2 points in Non-IID setting.

C. Robustness Analysis

1) *Non-IID Degree*: To demonstrate the resilience of PFL-ALP in varying degrees of Non-IID data distribution, we perform a comparison of MA and ASR in PFL-ALP, FedAvg, and multiple defense methods against ECBA on the CIFAR-10 dataset. The experimental results are depicted in Fig. 6, with the one class-expert distribution probability p ranging from 0.1 to 0.5, larger p signifies higher Non-IID degree.

Specifically, for server-side defenses, as p increases, the MA of FLTrust and FLAME drifts, whereas Multi-metrics shows non-monotonic variability. Their ASR correspondingly shows spikes and trend reversal, reflecting instability under increasing heterogeneity. For example, the ASR of FLTrust spikes at $p = 0.2$ with value around 19%, dips at $p = 0.3$ with value around 11%, and then increases again as p grows, while its MA steadily decreases to about 46% by $p = 0.5$. This instability is consistent with the shortages that when benign

TABLE V
ROBUSTNESS OF PFL-ALP AGAINST DIVERSE ATTACKS ON CIFAR-10 WITH MALICIOUS RATE OF 50%, 70% AND 90% (%)

Attack		50%				70%				90%			
		FedAvg [44]		PFL-ALP		FedAvg [44]		PFL-ALP		FedAvg [44]		PFL-ALP	
		MA $\uparrow$	ASR $\downarrow$	MA $\uparrow$	ASR $\downarrow$	MA $\uparrow$	ASR $\downarrow$	MA $\uparrow$	ASR $\downarrow$	MA $\uparrow$	ASR $\downarrow$	MA $\uparrow$	ASR $\downarrow$
SA	Wall	73.17	100	72.22(-0.95)	3.13(-96.87)	73.95	100	72.84(-1.11)	0.00(-100)	74.96	100	72.74(-2.22)	1.00(-99)
	Stripe	73.88	100	72.91(-0.97)	2.67(-97.33)	74.84	100	72.05(-2.79)	0.00(-100)	74.37	100	74.03(-0.34)	0.00(-100)
	Car	72.50	100	72.36(-0.14)	1.87(-98.13)	73.83	100	71.33(-2.50)	0.00(-100)	74.72	100	70.23(-4.49)	1.00(-99)
BA	3×3	85.32	91.32	74.52(-10.80)	2.44(-88.88)	86.36	93.32	75.30(-11.06)	2.32(-91)	87.22	94.77	77.11(-10.11)	2.23(-92.54)
	5×5	85.61	94.56	75.04(-10.57)	4.09(-90.47)	86.54	96.73	75.95(-10.59)	3.87(-92.86)	87.18	96.68	77.66(-9.52)	3.18(-93.50)
	7×7	85.46	95.71	74.46(-11)	3.67(-92.04)	86.71	97.71	75.89(-10.82)	3.80(-93.91)	86.63	98.43	77.77(-8.86)	3.54(-94.89)
ECBA		64.32	93.37	64.13(-0.19)	2.42(-90.95)	67.38	94.89	65.84(-1.54)	3.42(-91.47)	70.72	100	61.65(-9.07)	5.05(-94.95)
DBA		73.36	98.26	61.02(-12.34)	14.77(-83.49)	75.27	99.07	64.04(-11.23)	12.52(-86.55)	73.32	98.65	63.74(-9.58)	10.65(-88)
TLFA		64.21	95.10	71.01(+6.80)	9.32(-85.78)	64.22	85.90	67.72(+3.50)	12.38(-73.52)	62.91	93.33	68.74(+5.83)	23.93(-69.40)

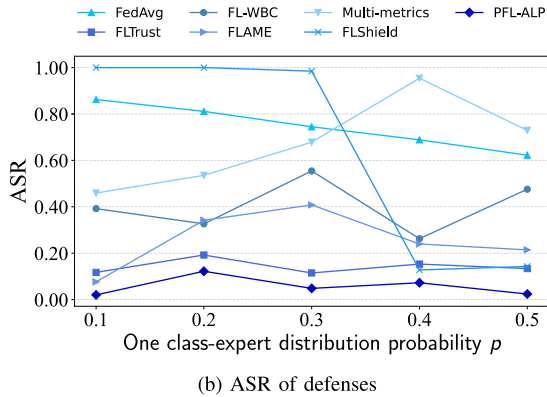
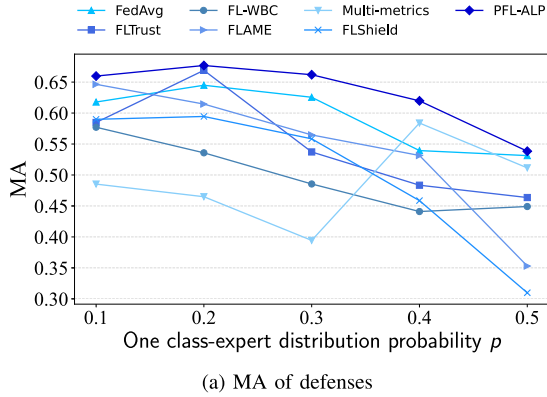


Fig. 6. MA and ASR of defenses against ECBA on CIFAR-10 in Non-IID.

updates naturally disperse under stronger heterogeneity, the outlier removal or consensus criteria used at the server become less robust.

For client-side defense, as p increases, the MA of FL-WBC erodes progressively, falling to around 45% at $p = 0.5$. Its ASR curve is also unstable, rising to about 55% at $p = 0.3$ and remaining elevated near 48% at $p = 0.5$.

In contrast, PFL-ALP demonstrates a better performance across all levels of heterogeneity. Its ASR remains uniformly

low, never exceeding 12%, and its MA is consistently the highest among all methods.

2) *Malicious Rate*: To demonstrate the robustness of PFL-ALP under varying levels of malicious rates, we perform a comparative analysis of PFL-ALP and FedAvg against different attacks in the CIFAR-10 dataset. Based on the previous analysis of a 30% malicious rate, we further explore the impact of malicious rates in 50%, 70%, and 90%. The experimental results are presented in Table V. It is shown that, since PFL-ALP performs client side purification to eliminate the backdoor information, its performance is minimally affected by the changes of malicious rate. Both MA and ASR have only slightly fluctuation, the varying is lower than 5% in most cases. For instance, when defending against SA, PFL-ALP consistently reduces the ASR by over 97% and sustains a comparable MA to FedAvg across all backdoor triggers. This underscores the remarkable resilience and effectiveness of PFL-ALP in mitigating attacks and maintaining performance stability.

D. PFL-ALP Against Advanced Backdoor Attacks

To evaluate the robustness of PFL-ALP against attackers with knowledge of the defense process, we further consider two advanced variations of DBA, including a saliency-guided Stealth-DBA (SDBA) and a gradient-aligned Adaptive-DBA (ADBA). The evaluation is conducted under both IID and Non-IID settings on three datasets and the results are shown in Table VI.

In SDBA, the trigger is placed inside the second-most salient region of the images, so it overlaps task-relevant features while leaving the most critical region untouched to balance the backdoor injection and main feature preservation. This attack poses challenges for the purification method, since when purifying activation to break the backdoor correlation, it necessarily pushes on the same regions that contain genuine semantics. The experimental results show that, under FedAvg, this placement leads to a decrease in MA, for example, around 2%-points on CIFAR-10 dataset in IID setting, and about 3%-points under Non-IID condition, while ASR remains high. In contrast, the ASR decreases markedly when PFL-ALP is employed. For example, on the CIFAR-10 dataset, the ASR

TABLE VI
PERFORMANCE COMPARISON OF FEDAVG AND PFL-ALP
UNDER ADVANCED DBA

Dataset	Attack	IID				Non-IID			
		FedAvg		PFL-ALP		FedAvg		PFL-ALP	
		MA $\uparrow$	ASR $\downarrow$	MA $\uparrow$	ASR $\downarrow$	MA $\uparrow$	ASR $\downarrow$	MA $\uparrow$	ASR $\downarrow$
CIFAR-10	DBA	73.69	98.27	62.33	13.86	54.30	91.21	41.06	6.77
	SDBA	71.74	99.96	61.91	26.66	51.35	99.07	39.89	3.32
	ADBA	71.47	98.08	61.12	18.13	62.93	96.54	46.47	9.22
Fashion-MNIST	DBA	99.98	88.40	81.93	8.04	87.97	99.98	70.86	11.62
	SDBA	87.54	97.29	79.66	23.43	84.81	98.29	66.56	11.68
	ADBA	87.73	99.91	77.21	9.06	85.25	99.96	68.72	15.95
MNIST	DBA	99.04	99.93	95.33	3.35	98.94	99.93	94.06	6.08
	SDBA	98.83	98.14	94.61	13.33	98.75	99.18	93.81	13.40
	ADBA	98.85	99.75	94.70	5.07	98.93	99.87	94.09	4.38

drops from 99.96% to 26.66% in the IID setting and from 99.07% to 3.32% in the Non-IID setting. While the MA has a modest decrease like FedAvg. The results indicate that PFL-ALP is powerful in purifying the backdoor impact of SDBA, while the loss of MA stems from the overlapping of the trigger with task-relevant features that directly support the class decision.

Instead of changing where the trigger is placed, ADDBA crafts model updates to evade cosine similarity-based filtering by optimizing a backdoor objective while aligning its gradients to a benign reference. PFL-ALP is resilient to this attack because it leverages the cosine similarity-based metric for reducing Non-IID bias rather than for backdoor defense. As a result, the ASR of PFL-ALP remains low in both IID and Non-IID conditions, coupled with compatible MA. For example, on MNIST dataset, the ASR drops from 99.75% in the IID condition and 99.87% in the Non-IID condition to 5.07% and 4.38%, respectively, and the MA is comparable to that under standard DBA.

These observations also clarify the strengths and limitations of our method. PFL-ALP extends the effectiveness of NAD for backdoor defense to Non-IID settings even against saliency-guided and gradient-aligned attacks. However, maintaining a high MA remains challenging, especially when triggers deliberately occupy salient content. This reflects an inherent limitation of local purification that mixed activations encoding both semantics and trigger may be over-suppressed.

E. Ablation Analysis

1) *Impact of β* : The hyperparameter β plays a crucial role in balancing the main task loss and NAD loss in local purification. A higher value of β signifies a greater emphasis on backdoor cleansing, potentially at the expense of main task accuracy. We present the MA and ASR of PFL-ALP against BA with triggers of various sizes under different β value in Fig. 7. The results indicate that as β increases, MA and ASR both decrease. Remarkably, the optimal value of β is identified as 1500, as it effectively reduces ASR while maintaining high MA levels. Further increasing the value of β is unlikely to evidently mitigate the ASR, yet it will inevitably lead to a reduction in MA.

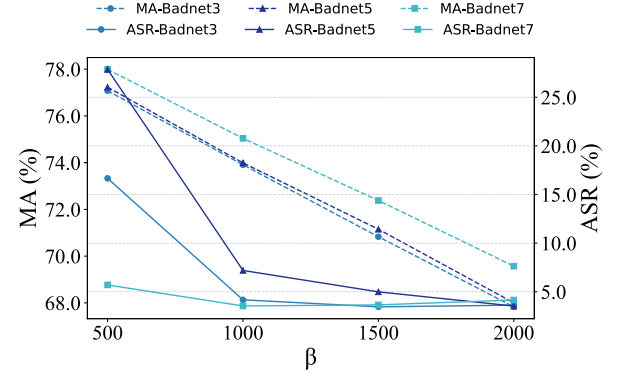


Fig. 7. MA and ASR of PFL-ALP against BA on CIFAR-10 under different β value.

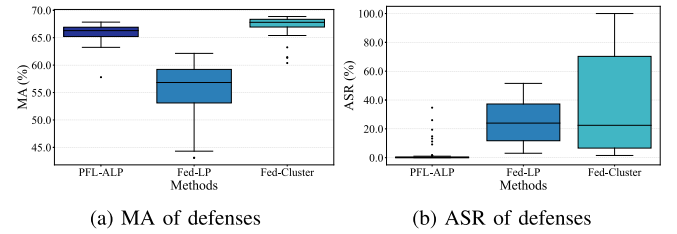


Fig. 8. MA and ASR of PFL-ALP against ECBA with different components. Fed-LP is PFL-ALP with only the local purification component, Fed-Cluster is PFL-ALP with only the clustering-based representative model generation.

2) *Component Contributions*: PFL-ALP comprises two main components: dynamic clustering-based representative model generation and attention-based local purification. The former focuses on generating representative models for clients with similar data distributions, mitigating the bias introduced by Non-IID settings. The second component aims to eliminate the backdoor impact that remains in the representative model.

We perform an ablation study on these two components to explain the effectiveness of PFL-ALP against backdoor attacks. Fig. 8 shows the MA and ASR of PFL-ALP and PFL-ALP with only one component against ECBA.

The results reveal that with only the dynamic clustering, each benign client achieves the highest MA but experiences significantly elevated ASR due to residual backdoor impact on representative models. Alternatively, employing only local purification results in reduced ASR for benign clients, with the cost of MA. By combining these two components, PFL-ALP effectively reduces ASR to nearly 0 while maintaining MA levels slightly below Fed-Cluster. This underscores the comprehensive effectiveness of PFL-ALP in mitigating backdoor attacks.

F. Visualization of PFL-ALP

Fig. 9 presents a comparison of the attention maps derived from the trained models of FedAvg, FLTrust, FL-WBC, FLAME, Multi-metrics, FLShield and our method PFL-ALP on three images backdoored by BA with 3×3 pixel block trigger. The visualization results reveal that the attention of model trained by FedAvg shifts to the backdoor trigger located

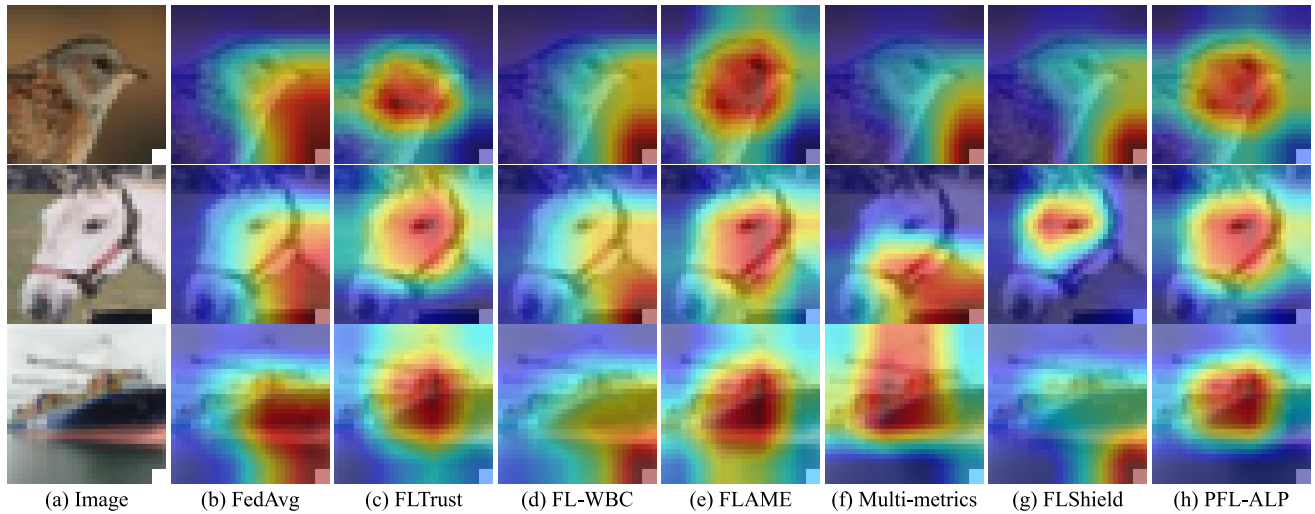


Fig. 9. Attention maps of different models. Column (a) is the images with backdoor trigger, columns (b)-(h) are the attention maps of models trained by various backdoor defense methods.

at the lower right corner, resulting in misclassification. In column (g), FLShield exhibits insufficient efficacy, as attention tends to tightly align with the trigger, leading to high ASR. In columns (d) and (f), both FL-WBC and Multi-metrics provide insufficient backdoor trigger suppression, since the models produce large and diffuse attention that cover both the object and the trigger, which erodes semantic focus and leads to high ASR. As shown in column (e), while FLAME successfully suppresses the backdoor trigger, the model attention often fragments or shifts off the main object. In contrast, the model trained by PFL-ALP and FLTrust successfully suppresses the trigger while preserving attention concentrated on the task-relevant object, thereby achieving the lowest ASR and better performance on the main task.

VI. CONCLUSION

In this paper, we have proposed PFL-ALP, a hybrid defense mechanism that performs server-side dynamic clustering to generate representative models that mitigates the bias introduced by the Non-IID scenario and further implements client-side attention-based purification to cleanse the backdoor information in the representative models. We establish a convergence guarantee for PFL-ALP and validate its effectiveness through comprehensive experiments against multiple defenses. Across various backdoor settings, PFL-ALP achieves superior robustness in both IID and Non-IID scenarios, remaining effective even when the malicious client rate reaches 90%. Furthermore, PFL-ALP is compatible with STPC, reducing the risk of inference attacks without altering the defense pipeline.

However, while PFL-ALP extends the effectiveness of NAD-based backdoor purification to Non-IID scenarios, the MA is hard to maintain because the purification may over-suppress activations mixing semantic and trigger features. In future work, our focus will be on enhancing local purification techniques by exploring methods such as semantics-aware purification, attention-contrastive learning, and knowledge dis-

tillation to effectively cleanse the backdoor impacts, while maintaining the main task accuracy.

REFERENCES

- [1] Q. Yang, Y. Liu, T. Chen, and Y. Tong, "Federated machine learning: Concept and applications," *ACM Trans. Intell. Syst. Technol.*, vol. 10, no. 2, pp. 1–19, 2019.
- [2] Y. Wan, Y. Qu, W. Ni, Y. Xiang, L. Gao, and E. Hossain, "Data and model poisoning backdoor attacks on wireless federated learning, and the defense mechanisms: A comprehensive survey," *IEEE Commun. Surveys Tuts.*, vol. 26, no. 3, pp. 1861–1897, 3rd Quart., 2024.
- [3] Y. Chai et al., "MalFSCIL: A few-shot class-incremental learning approach for malware detection," *IEEE Trans. Inf. Forensics Security*, vol. 20, pp. 2999–3014, 2025.
- [4] T. D. Nguyen, T. Nguyen, P. L. Nguyen, H. H. Pham, K. D. Doan, and K.-S. Wong, "Backdoor attacks and defenses in federated learning: Survey, challenges and future research directions," *Eng. Appl. Artif. Intell.*, vol. 127, Jan. 2024, Art. no. 107166.
- [5] E. Bagdasaryan, A. Veit, Y. Hua, D. Estrin, and V. Shmatikov, "How to backdoor federated learning," in *Proc. Int. Conf. Artif. Intell. Statist.*, 2020, pp. 2938–2948.
- [6] T. Gu, B. Dolan-Gavitt, and S. Garg, "BadNets: Identifying vulnerabilities in the machine learning model supply chain," 2017, *arXiv:1708.06733*.
- [7] C. Xie, K. Huang, P. Chen, and B. Li, "DBA: Distributed backdoor attacks against federated learning," in *Proc. Int. Conf. Learn. Represent.*, 2020, pp. 1–19.
- [8] K. Wei et al., "Federated learning with differential privacy: Algorithms and performance analysis," *IEEE Trans. Inf. Forensics Security*, vol. 15, pp. 3454–3469, 2020.
- [9] J. Geiping, H. Bauermeister, H. Dröge, and M. Moeller, "Inverting gradients—How easy is it to break privacy in federated learning?," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 16937–16947.
- [10] L. Melis, C. Song, E. De Cristofaro, and V. Shmatikov, "Exploiting unintended feature leakage in collaborative learning," in *Proc. IEEE Symp. Secur. Privacy (SP)*, May 2019, pp. 691–706.
- [11] P. Blanchard, E. M. E. Mhamdi, R. Guerraoui, and J. Stainer, "Machine learning with adversaries: Byzantine tolerant gradient descent," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 118–128.
- [12] D. Yin, Y. Chen, K. Ramchandran, and P. L. Bartlett, "Byzantine-robust distributed learning: Towards optimal statistical rates," in *Proc. Int. Conf. Mach. Learn.*, 2018, pp. 5650–5659.
- [13] Z. Zhang, X. Cao, J. Jia, and N. Z. Gong, "FLDetector: Defending federated learning against model poisoning attacks via detecting malicious clients," in *Proc. 28th ACM SIGKDD Conf. Knowl. Discovery Data Mining*, Aug. 2022, pp. 2545–2555.

- [14] M. Vucovich, D. Quinn, K. Choi, C. Redino, A. Rahman, and E. Bowen, "FedBayes: A zero-trust federated learning aggregation to defend against adversarial attacks," in *Proc. IEEE 14th Annu. Comput. Commun. Workshop Conf. (CCWC)*, Jan. 2024, pp. 0028–0035.
- [15] X. Cao, M. Fang, J. Liu, and N. Z. Gong, "FLTrust: Byzantine-robust federated learning via trust bootstrapping," 2020, *arXiv:2012.13995*.
- [16] J. Zhang et al., "BadCleaner: Defending backdoor attacks in federated learning via attention-based multi-teacher distillation," *IEEE Trans. Dependable Secure Comput.*, vol. 21, no. 5, pp. 4559–4573, Sep. 2024.
- [17] J. Sun, A. Li, L. DiValentin, A. Hassanzadeh, Y. Chen, and H. Li, "FL-WBC: Enhancing robustness against model poisoning attacks in federated learning from a client perspective," in *Proc. Adv. Neural Inf. Process. Syst.*, 2021, pp. 12613–12624.
- [18] Y. Cai, J. Qiu, F. Zhang, Q. Li, and L. Chen, "A knowledge extraction framework on cyber threat reports with enhanced security profiles," in *Proc. 48th Int. ACM SIGIR Conf. Res. Develop. Inf. Retr.*, Jul. 2025, pp. 326–336.
- [19] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," 2015, *arXiv:1503.02531*.
- [20] H. Ma, S. Qi, J. Yao, Y. Yuan, Y. Zou, and D. Yu, "Fed-NAD: Backdoor resilient federated learning via neural attention distillation," in *Proc. 10th IEEE Int. Conf. Intell. Data Secur. (IDS)*, May 2024, pp. 7–13.
- [21] Z. Liu, W. He, C.-H. Chang, J. Ye, H. Li, and X. Li, "SPFL: A self-purified federated learning method against poisoning attacks," *IEEE Trans. Inf. Forensics Security*, vol. 19, pp. 6604–6619, 2024.
- [22] S. Huang et al., "Scope: On detecting constrained backdoor attacks in federated learning," *IEEE Trans. Inf. Forensics Security*, vol. 20, pp. 3302–3315, 2025.
- [23] P. Fang and J. Chen, "On the vulnerability of backdoor defenses for federated learning," in *Proc. AAAI Conf. Artif. Intell.*, 2023, vol. 37, no. 10, pp. 11800–11808.
- [24] Y. Li, X. Lyu, N. Koren, L. Lyu, B. Li, and X. Ma, "Neural attention distillation: Erasing backdoor triggers from deep neural networks," in *Proc. Int. Conf. Learn. Represent.*, 2021, pp. 1–19.
- [25] D. Demmler, T. Schneider, and M. Zohner, "ABY—A framework for efficient mixed-protocol secure two-party computation," in *Proc. Netw. Distrib. Syst. Secur. Symp.*, 2015, pp. 1–15.
- [26] H. Wang et al., "Attack of the tails: Yes, you really can backdoor federated learning," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, pp. 16070–16084.
- [27] V. Tolpegin, S. Truex, M. E. Gürsoy, and L. Liu, "Data poisoning attacks against federated learning systems," in *Proc. 25th Eur. Symp. Res. Comput. Secur.*, Sep. 2020, pp. 480–501.
- [28] H. Li et al., "3DFed: Adaptive and extensible framework for covert backdoor attack in federated learning," in *Proc. IEEE Symp. Secur. Privacy (SP)*, May 2023, pp. 1893–1907.
- [29] Z. Zhang et al., "Neurotoxin: Durable backdoors in federated learning," in *Proc. Int. Conf. Mach. Learn.*, 2022, pp. 26429–26446.
- [30] T. D. Nguyen et al., "FLAME: Taming backdoors in federated learning," in *Proc. 31st USENIX Secur. Symp. (USENIX Secur.)*, 2022, pp. 1415–1432.
- [31] R. J. G. B. Campello, D. Moulavi, and J. Sander, "Density-based clustering based on hierarchical density estimates," in *Proc. Pacific-Asia Conf. Knowl. Discovery Data Mining*, vol. 7819, 2013, pp. 160–172.
- [32] X. Mu et al., "FedDMC: Efficient and robust federated learning via detecting malicious clients," *IEEE Trans. Dependable Secure Comput.*, vol. 21, no. 6, pp. 5259–5274, Nov. 2024.
- [33] S. Huang, Y. Li, C. Chen, L. Shi, and Y. Gao, "Multi-metrics adaptively identifies backdoors in federated learning," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 4629–4639.
- [34] M. Fang, X. Cao, J. Jia, and N. Z. Gong, "Local model poisoning attacks to Byzantine-robust federated learning," in *Proc. 29th USENIX Secur. Symp. (USENIX Secur. 20)*, 2019, pp. 1605–1622.
- [35] S. P. Karimireddy, L. He, and M. Jaggi, "Learning from history for Byzantine robust optimization," in *Proc. Int. Conf. Mach. Learn. (ICML)*, vol. 139, 2021, pp. 5311–5319.
- [36] V. Shejwalkar and A. Houmansadr, "Manipulating the Byzantine: Optimizing model poisoning attacks and defenses for federated learning," in *Proc. Netw. Distrib. Syst. Secur. Symp.*, 2021, pp. 1148–1164.
- [37] J. Le, D. Zhang, X. Lei, L. Jiao, K. Zeng, and X. Liao, "Privacy-preserving federated learning with malicious clients and honest-but-curious servers," *IEEE Trans. Inf. Forensics Security*, vol. 18, pp. 4329–4344, 2023.
- [38] M. Hao, H. Li, G. Xu, H. Chen, and T. Zhang, "Efficient, private and robust federated learning," in *Proc. Annu. Comput. Security Appl. Conf.*, 2021, pp. 45–60.
- [39] C.-L. Chen, L. Golubchik, and M. Paolieri, "Backdoor attacks on federated meta-learning," 2020, *arXiv:2006.07026*.
- [40] O. Vinyals, C. Blundell, T. Lillicrap, K. Kavukcuoglu, and D. Wierstra, "Matching networks for one shot learning," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 29, 2016, pp. 3637–3645.
- [41] E. Kabir, Z. Song, M. R. Ur Rashid, and S. Mehnaz, "FLShield: A validation based federated learning framework to defend against poisoning attacks," in *Proc. IEEE Symp. Secur. Privacy (SP)*, May 2024, pp. 2572–2590.
- [42] J. Zhang, D. Wu, C. Liu, and B. Chen, "Defending poisoning attacks in federated learning via adversarial training method," in *Proc. 3rd Int. Conf. Frontiers Cyber Secur.*, Nov. 2020, pp. 83–94.
- [43] Y. Jiang, W. Zhang, and Y. Chen, "Data quality detection mechanism against label flipping attacks in federated learning," *IEEE Trans. Inf. Forensics Security*, vol. 18, pp. 1625–1637, 2023.
- [44] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. Y. Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proc. 20th Int. Conf. Artif. Intell. Statist.*, 2017, pp. 1273–1282.
- [45] X. Li, Z. Qu, S. Zhao, B. Tang, Z. Lu, and Y. Liu, "LoMar: A local defense against poisoning attack on federated learning," *IEEE Trans. Dependable Secure Comput.*, vol. 20, no. 1, pp. 437–450, Jan. 2023.
- [46] M. Nwadike, T. Miyawaki, E. Sarkar, M. Maniatakos, and F. Shamout, "Explainability matters: Backdoor attacks on medical imaging," 2020, *arXiv:2101.00008*.
- [47] P. J. Rousseeuw, "Silhouettes: A graphical aid to the interpretation and validation of cluster analysis," *J. Comput. Appl. Math.*, vol. 20, pp. 53–65, Nov. 1987.
- [48] Z. Zhang et al., "LSFL: A lightweight and secure federated learning scheme for edge computing," *IEEE Trans. Inf. Forensics Security*, vol. 18, pp. 365–379, 2023.
- [49] X. Li, X. Yang, Z. Zhou, and R. Lu, "Efficiently achieving privacy preservation and poisoning attack resistance in federated learning," *IEEE Trans. Inf. Forensics Security*, vol. 19, pp. 4358–4373, 2024.
- [50] X. Li, K. Huang, W. Yang, S. Wang, and Z. Zhang, "On the convergence of FedAvg on non-IID data," in *Proc. Int. Conf. Learn. Represent.*, 2019, pp. 1–26.
- [51] L. Tan et al., "AdaFed: Optimizing participation-aware federated learning with adaptive aggregation weights," *IEEE Trans. Netw. Sci. Eng.*, vol. 9, no. 4, pp. 2708–2720, Jul. 2022.
- [52] P. Guo et al., "A new federated learning framework against gradient inversion attacks," in *Proc. AAAI Conf. Artif. Intell.*, 2024, pp. 16969–16977.
- [53] L. Wang, M. Huang, Z. Zhang, M. Li, J. Wang, and K. Gai, "RaSA: Robust and adaptive secure aggregation for edge-assisted hierarchical federated learning," *IEEE Trans. Inf. Forensics Security*, vol. 20, pp. 4280–4295, 2025.
- [54] Y. Lecun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," *Proc. IEEE*, vol. 86, no. 11, pp. 2278–2324, 1998.
- [55] H. Xiao, K. Rasul, and R. Vollgraf, "Fashion-MNIST: A novel image dataset for benchmarking machine learning algorithms," 2017, *arXiv:1708.07747*.
- [56] A. Krizhevsky and G. Hinton, "Learning multiple layers of features from tiny images," Dept. Comput. Sci., University of Toronto, Toronto, ON, Canada, Tech. Rep. 4, 2009.
- [57] N. Ye et al., "OoD-bench: Quantifying and understanding two dimensions of out-of-distribution generalization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 7937–7948.



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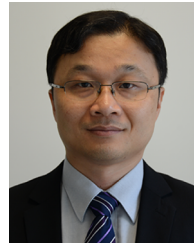


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